



Measure Difficult Angles With This **Digital Protractor**

Ashwini Kumar Sinha

Digital compass and protractor are two of the most basic tools for specifying directions and measuring angles. But sometimes it is difficult to get accurate angle measurement for certain structures and geometrical shapes using these traditional tools, for example, the angle between two adjacent walls.

So I thought of developing a digital protractor to make the work a bit easier. I came up with two different methods to make a digital protractor: first by converting the MPU6050 sensor data into angle, and second by converting stepper motor rotation into angle. The second method is explained here.

For this project, we need the following components:

- Arduino Pro Mini
- ULN2003A module
- 28BYJ stepper motor
- Two pushbutton switches
- OLED display (I2C interface)
- 5V battery
- Jumper wires

The circuit diagram of digital protractor using stepper motor is shown in Fig. 1. All the positive terminals (+) should be connected to 5V supply and negative terminals (-) to common

ground (GND). Switch S1 is used for rotating stepper motor in clockwise (CW) direction and switch S2 for counter-clockwise (CCW) direction. A five-wire, unipolar stepper motor (28BYJ) is connected across the ULN2003A motor driver module.

Arduino Pro Mini connection details with the components are given under Table 1.

Coding

The firmware (digital_protractor.ino) for this project is written in Arduino programming language. First, add relevant software library to Arduino IDE from Include Library as shown in Fig. 2.

Install the '28BYJ_48' libraries as shown in Fig. 3. After successful

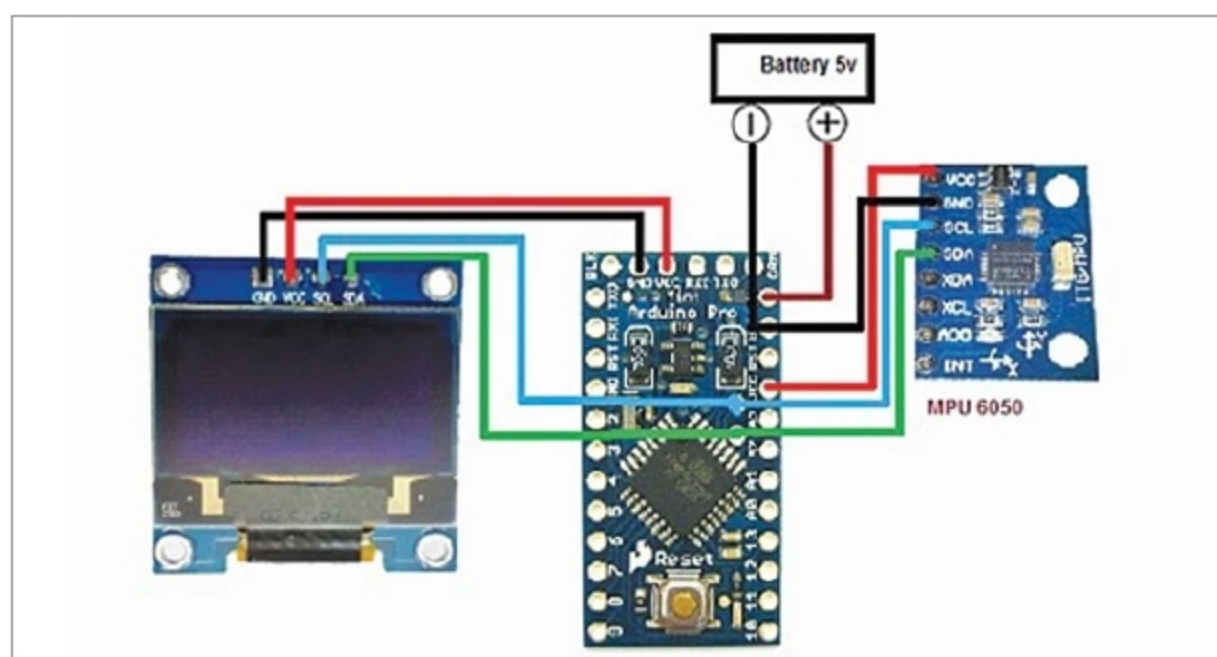


Fig. 1: Circuit diagram of digital protractor using stepper motor

Arduino Connections for Digital Protractor

Arduino	Component
SCL	SCL pin of OLED
SDA	SDA pin of OLED
GND	GND pin of OLED
5V	VCC pin of OLED
PIN8	IN1 of ULN2003A
PIN9	IN2 of ULN2003A
PIN10	IN3 of ULN2003A
PIN11	IN4 of ULN2003A
PIN12	S2 PIN2
PIN13	S1 PIN2
5V	Pin1 of S1 and S2

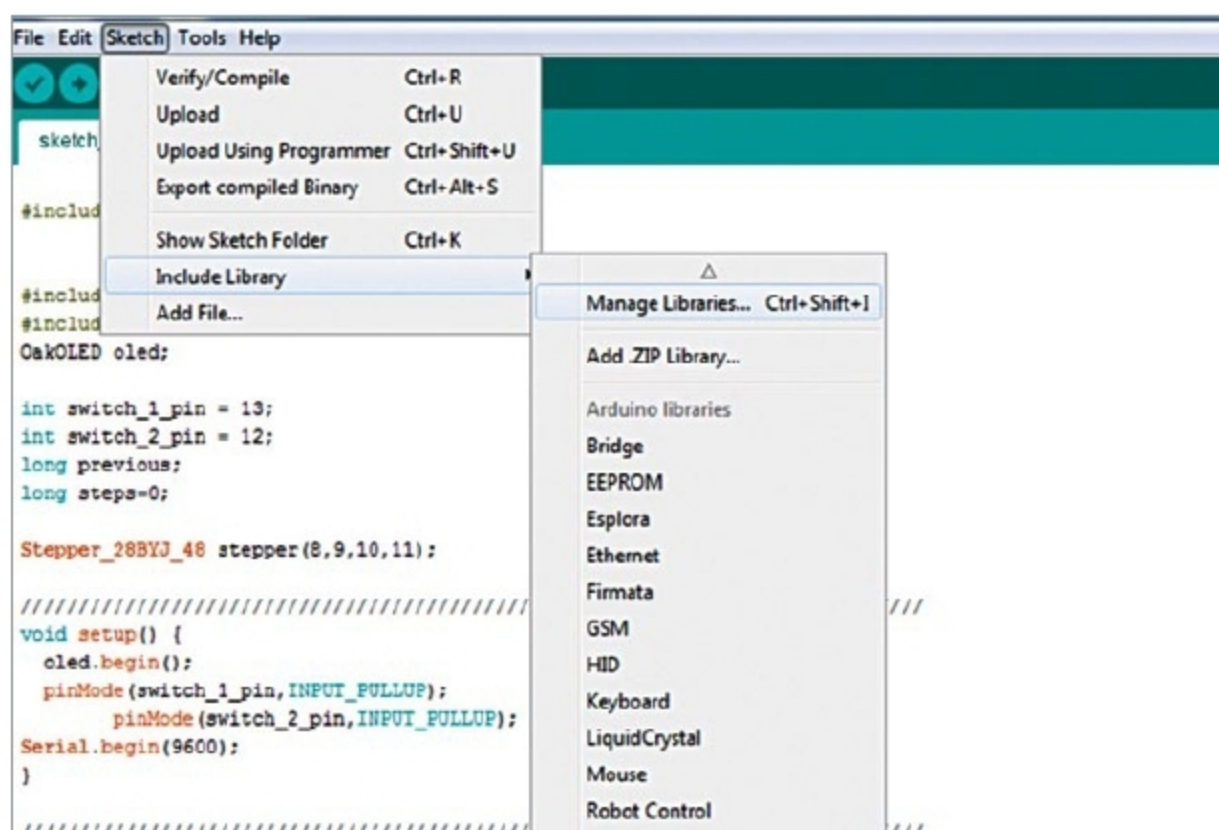


Fig. 2: Include Library in Arduino IDE